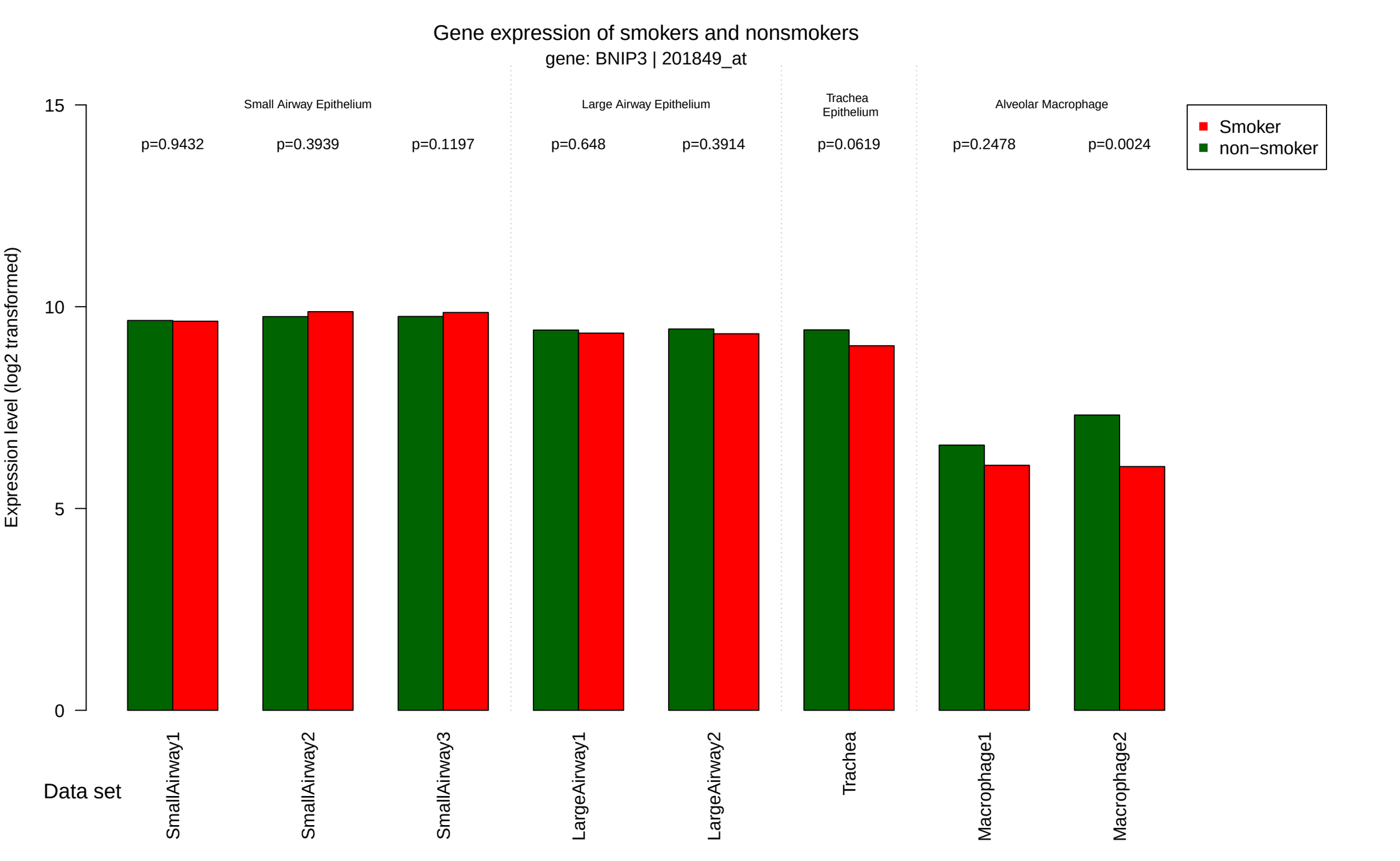
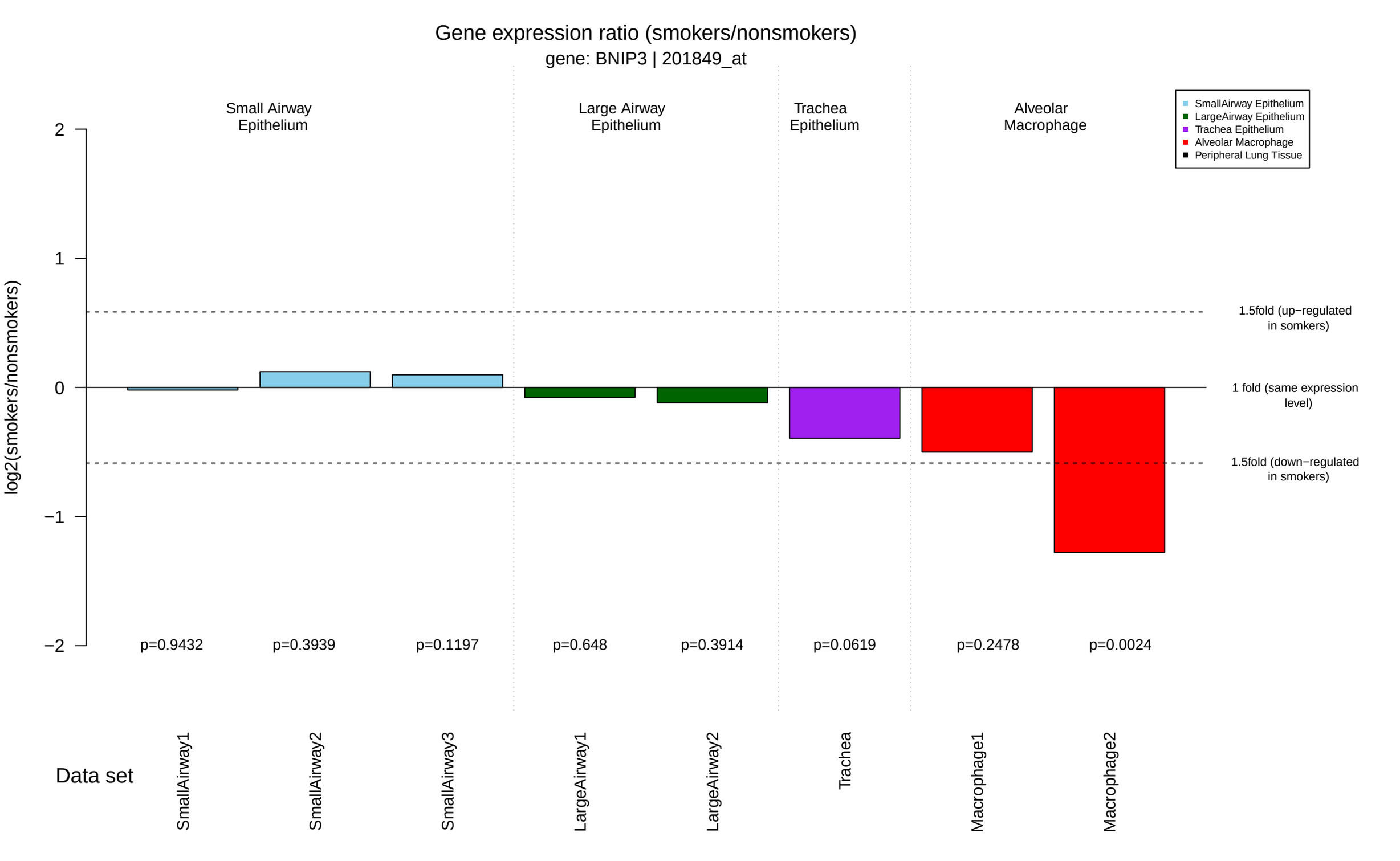
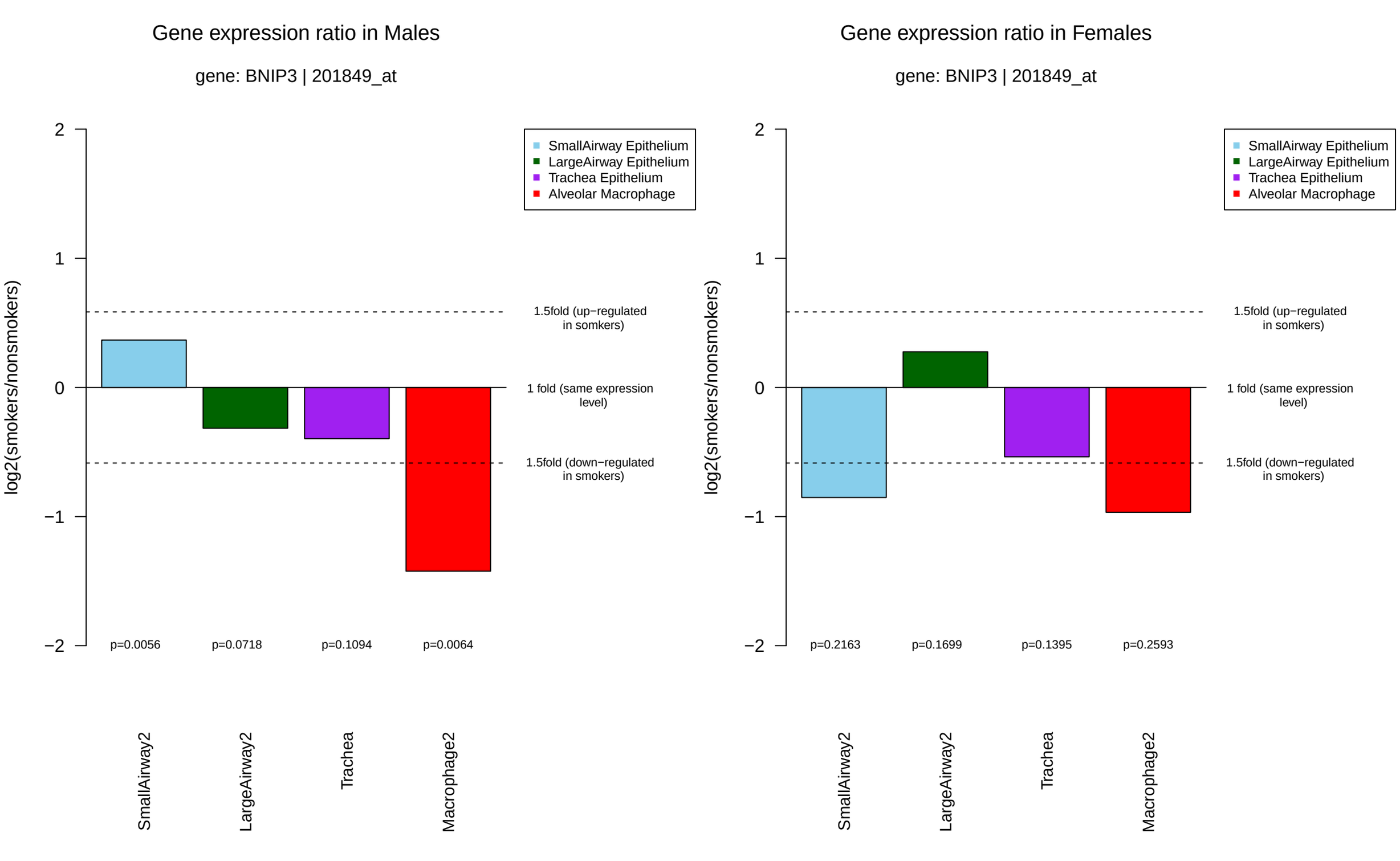
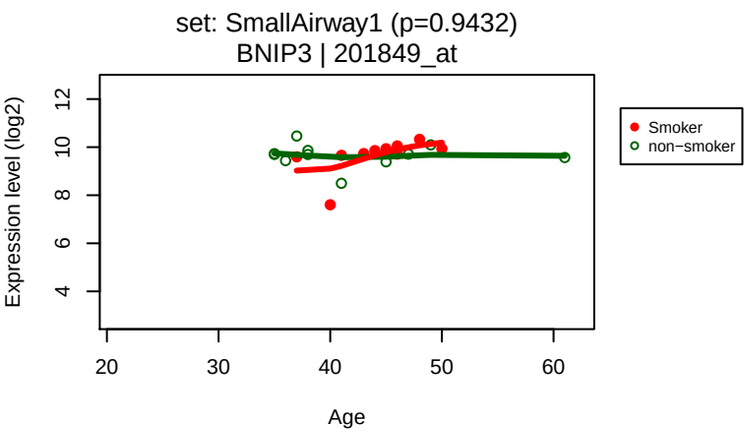
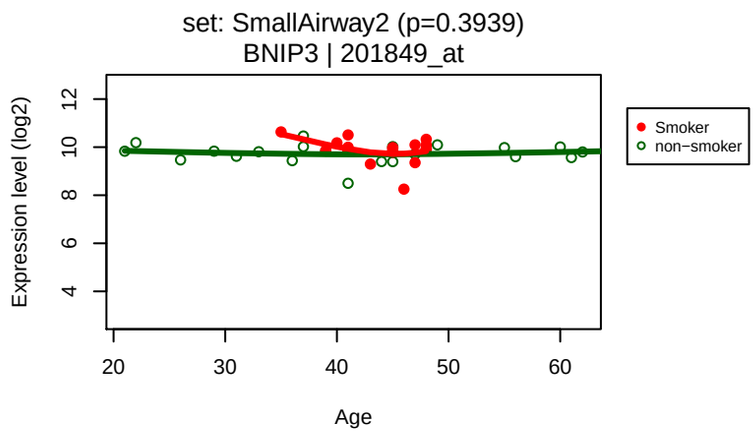
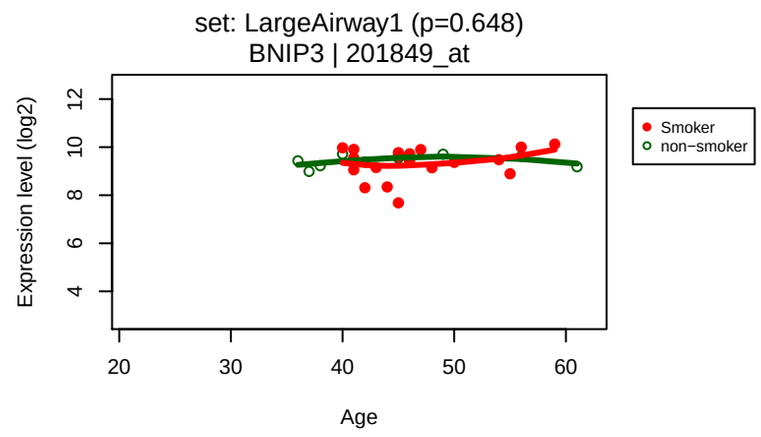

Gene expression of smokers and nonsmokers
gene: BNIP3 | 201849_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9432
p=0.3939
p=0.1197
p=0.648
p=0.3914
p=0.0619
p=0.2478
p=0.0024
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

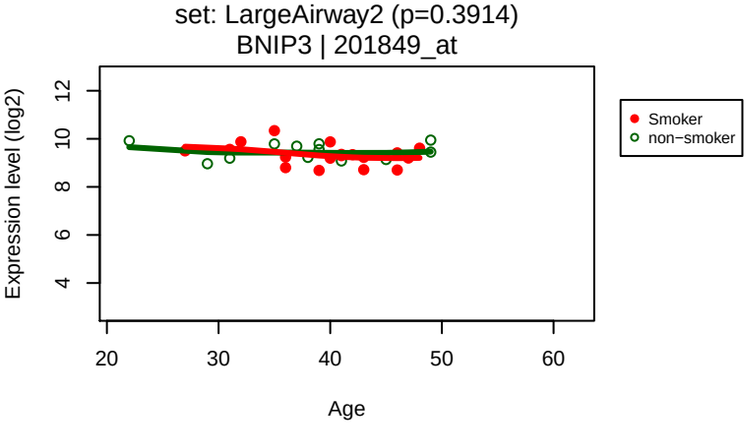

Gene expression ratio (smokers/nonsmokers)
gene: BNIP3 | 201849_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9432
p=0.3939
p=0.1197
p=0.648
p=0.3914
p=0.0619
p=0.2478
p=0.0024
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

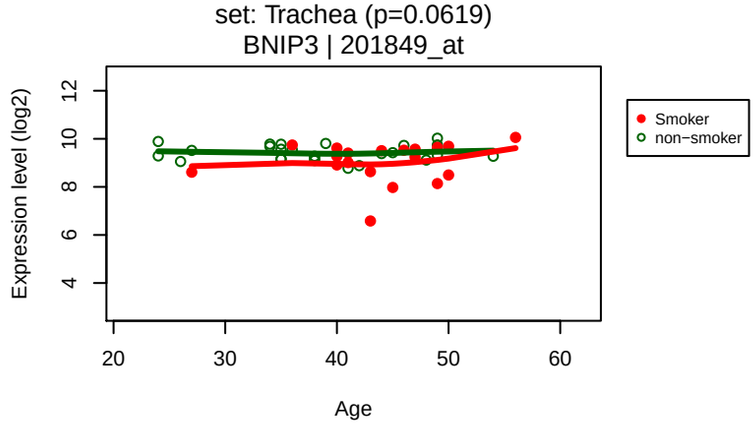

Gene expression ratio in Males
gene: BNIP3 | 201849_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0056
p=0.0718
p=0.1094
p=0.0064
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BNIP3 | 201849_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2163
p=0.1699
p=0.1395
p=0.2593
SmallAirway2
LargeAirway2
Trachea
Macrophage2

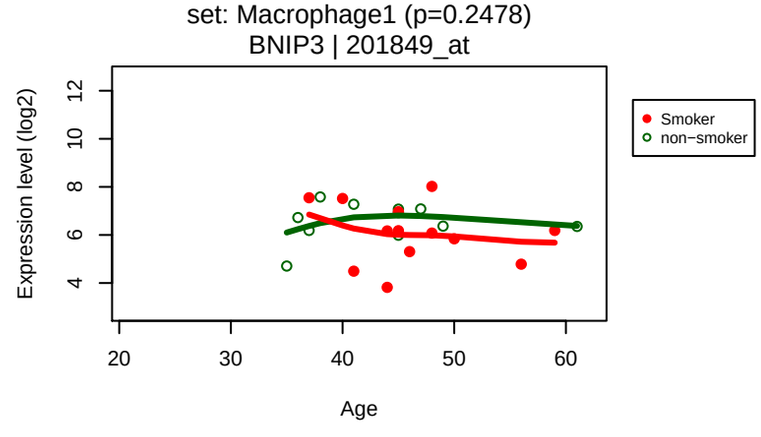

set: SmallAirway1 (p=0.9432)
BNIP3 | 201849_at
Expression level (log2)
Age
Smoker
non−smoker

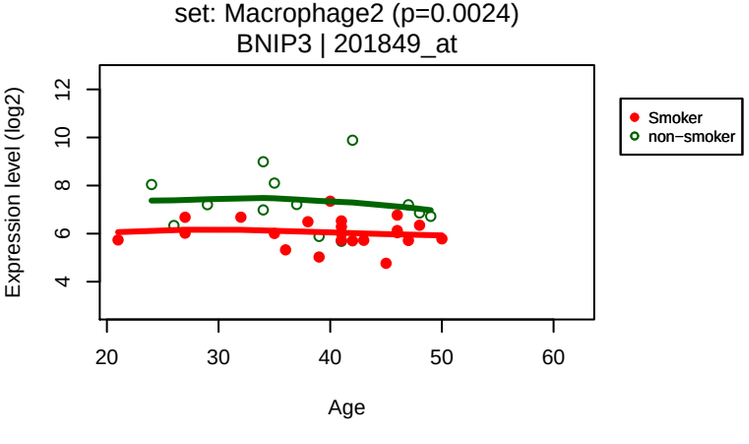

set: SmallAirway2 (p=0.3939)
BNIP3 | 201849_at
Expression level (log2)
Age
Smoker
non−smoker

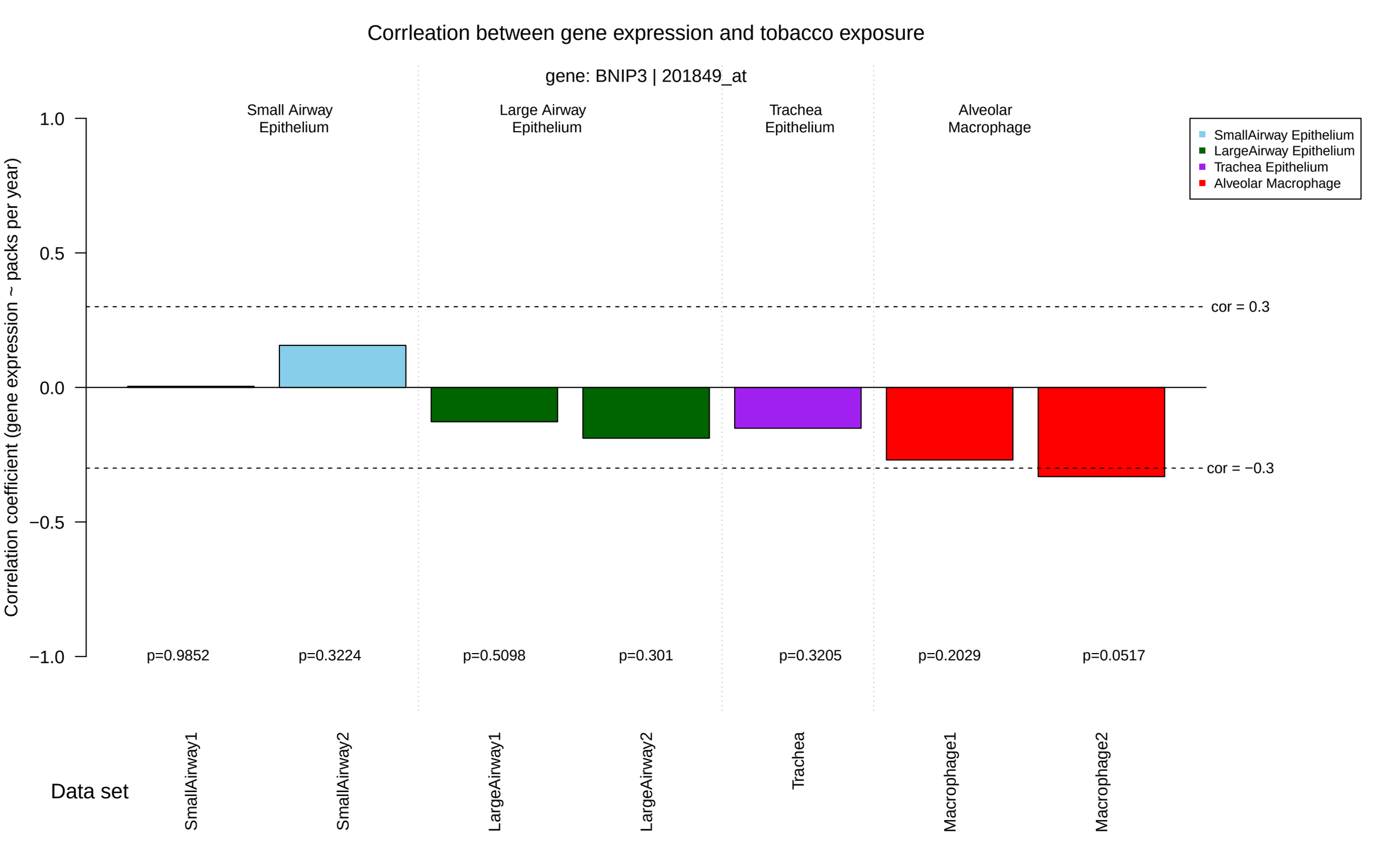

set: LargeAirway1 (p=0.648)
BNIP3 | 201849_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3914)
BNIP3 | 201849_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0619)
BNIP3 | 201849_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2478)
BNIP3 | 201849_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0024)
BNIP3 | 201849_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BNIP3 | 201849_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9852
p=0.3224
p=0.5098
p=0.301
p=0.3205
p=0.2029
p=0.0517
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2